Chapter 03
Numerical Initial Value Problem

Lesson 2

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3.2. Linear Multistep Methods

- In previous section we saw two methods (Euler and Trapezium) for approximating the solutions of certain initial value problems.

- In this section we will see that those two methods are special cases of a more general collection of techniques called linear multistep methods.

- Techniques for determining the properties of these methods will be presented.
- Another class of approximations, called Runge-Kutta methods, will also be discussed briefly.
- These are not linear multistep methods, but the two are sometimes used in conjunction.

3.2.1. General Linear Multistep Methods

The general k −step linear multistep method is given by

$$\boldsymbol{\alpha_k y_{n+k} + \cdots + \alpha_1 y_{n+1} + \alpha_0 y_n = h(\beta_k f_{n+\cdots+\beta_1 f_{n+1}+\beta_0 f_n+k})}$$

Or equivalently.

$$\boldsymbol{\sum_{j=0}^{k} \alpha_j y_{n+j} = h \sum_{j=0}^{k} \beta_j f_{n+j}}$$

It is always the case that $\alpha_k \neq 0$. Also, at least one of α_0 and β_0 will be non-zero.

A linear multistep method is defined by the choice of the quantities

$$k, \alpha_0, \alpha_1, \dots, \alpha_k, \beta_0, \beta_1, \dots, \beta_k$$

- If $\beta_k = 0$ the method is called explicit. (Because at each step, when we are trying to find the newest y_{n+k}, there is no appearance of this unknown on the right-hand side.)

- If $\beta_k \neq 0$ the method is called implicit. (Because y_{n+k} now appears on both sides of the equation (on the right-hand side it appears through

$f_{n+k} = f((n+k)h, y_{n+k})$, and we cannot, in general, rearrange to get an explicit formula for y_{n+k}.

Then you can see that

A linear multistep method uses **several previous values** of the numerical solution to calculate the next value.

Example 01

Write down the linear multistep scheme defined by the choices $k = 1, \alpha_0 = -1, \alpha_1 = 1, \beta_0 = \beta_1 = \frac{1}{2}$

Solution

Here $k = 1$ so that

$$\alpha_1 y_{n+1} + \alpha_0 y_n = h(\beta_1 f_{n+1} + \beta_0 f_n)$$

and substituting the values in for the four coefficients gives

$$y_{n+1} - y_n = h\left(\frac{1}{2} f_{n+1} + \frac{1}{2} f_n\right)$$

which, as we know, is the trapezium method.

Example 02

Write down the linear multistep scheme defined by the choices $k = 1$, $\alpha_0 = -1$, $\alpha_1 = 1$, $\beta_0 = 1$ and $\beta_1 = 0$.

Example 03

Write down the linear multistep scheme defined by the choices $k = 2$, $\alpha_0 = 0$, $\alpha_1 = -1$, $\alpha_2 = 1$, $\beta_2 = 0$, $\beta_1 = \frac{3}{2}$ and $\beta_0 = -\frac{1}{2}$.

Example 04

A numerical scheme has been used to approximate the solution of

$$\frac{dy}{dt} = t + y, \qquad y(0) = 3$$

and has produced the following estimates, to 6 decimal places,

$$y(0.4) \approx 4.509822, \quad y(0.45) \approx 4.755313$$

Now use the 2-step, explicit linear multistep scheme

$$y_{n+2} - y_{n+1} = h\,(1.5 f_{n+1} - 0.5 f_n)$$

to approximate $y(0.5)$.

Evidently the value $h = 0.05$ will serve our purposes and we seek $y_{10} \approx y(0.5)$. The values we will need to use in our implementation of the 2-step scheme are $y_9 = 4.755313$ and

$$f_9 = f(0.45, y_9) = 5.205313 \qquad f_8 = f(0.4, y_8) = 4.909822$$

to 6 decimal places since $f(t, y) = t + y$. It follows that

$$\begin{aligned} y_{10} &= y_9 + 0.05 \times (1.5 f_9 - 0.5 f_8) \\ &= 5.022966 \end{aligned}$$

And we conclude that $y(0.5) \approx 5.022966$, where this approximation has been given to 6 decimal places.

Example 05

A numerical scheme has been used to approximate the solution of

$$\frac{dy}{dt} = t/y \qquad y(0) = -2$$

and has produced the following estimates, to 6 decimal places,

$$y(0.24) \approx -2.013162, \quad y(0.26) \approx -2.015546$$

Now use the 2-step, explicit linear multistep scheme

$$y_{n+2} - \tfrac{1}{2}y_{n+1} - \tfrac{1}{2}y_n = \tfrac{3}{2}hf_{n+1}$$

to approximate $y(0.28)$.

Zero stability

Suppose we are solving an ODE numerically.

As an example:

Consider the ODE

$$\frac{dy}{dt} = y$$

with an initial condition

$$y(0) = 1.$$

The exact solution is

$$y = e^t$$

- Even if our initial values have **very small errors**, we want those errors to remain controlled as we move forward.
- For example:

$$\text{small error} \rightarrow \text{small error} \rightarrow \text{small error}$$

is desirable.

- But in many real problems, the exact solution is difficult or impossible to find.

If the errors behave like

$$0.001 \rightarrow 0.01 \rightarrow 0.1 \rightarrow 1 \rightarrow 10$$

then the numerical solution becomes unreliable.

So, we use **numerical methods** such as:

- Euler's method
- Runge–Kutta methods
- Linear multistep methods

to calculate approximate values such as

$$y(0.1), y(0.2), y(0.3), \ldots$$

According to that what does "zero stable" mean?

- Zero stability means that small errors in the starting values do not grow without bound as the computation proceeds.

- "A scheme that is zero stable will not produce approximations which grow unrealistically with t."

The main idea is:

- Zero stability prevents uncontrolled growth of numerical errors.

First characteristic polynomial

- For a linear multistep method, we define the **first characteristic polynomial** as,

$$\rho(z) = \alpha_0 + \alpha_1 z + \alpha_2 z^2 + \ldots \alpha_k z^k$$

where the α_i are the coefficients of the linear multistep method.

The important point is:

> We take the coefficients multiplying the y's and use them to construct a polynomial in z.

For example, suppose we have

$$y_{n+2} - 3y_{n+1} + 2y_n = h(\cdots).$$

The coefficients of the y-terms are

$$a_0 = 2, \qquad a_1 = -3, \qquad a_2 = 1.$$

Therefore,

$$\boxed{\rho(z) = 2 - 3z + z^2}$$

or

$$\boxed{\rho(z) = z^2 - 3z + 2}.$$

That is the **first characteristic polynomial**.

The linear multistep scheme

$$\sum_{j=0}^{k} \alpha_j \; y_{n+j} = h \sum_{j=0}^{k} \beta_j \; f_{n+j}.$$

is said to be **zero stable** if the zeros of the first characteristic polynomial are such that

1. none is larger than 1 in magnitude
2. any zero equal to 1 in magnitude is simple (that is, not repeated)

Example 01

Find the roots of the first characteristic polynomial for each of the examples below and determine whether or not the method is zero stable.

(a) $y_{n+1} - y_n = hf_n$

(b) $y_{n+1} - 2y_n = hf_n$

(c) $y_{n+2} + 3y_{n+1} - 4y_n = h\left(2f_{n+2} + f_{n+1} + 2f_n\right)$

(d) $y_{n+2} - y_{n+1} = \frac{3}{2}hf_{n+1}$

(e) $y_{n+2} - 2y_{n+1} + y_n = h(f_{n+2} - f_n)$

(a) In this case $\rho(z) = z - 1$ and the single zero of ρ is $z = 1$. This is a simple (that is, not repeated) root with magnitude equal to 1, so the method is zero stable.

(b) $\rho(z) = z - 2$ which has one zero, $z = 2$. This has magnitude $2 > 1$ and therefore the method is not zero stable.

(c) $\rho(z) = z^2 + 3z - 4 = (z - 1)(z + 4)$. One root is $z = -4$ which has magnitude greater than 1 and the method is therefore not zero stable.

(d) Here $\alpha_2 = 1$, $\alpha_1 = -1$ and $\alpha_0 = 0$, therefore

$$\rho(z) = z^2 - z = z(z - 1)$$

which has two zeros, $z = 0$ and $z = 1$. These both have magnitude less than or equal to 1 and there is no repeated zero with magnitude equal to 1, so the method is zero stable.

(e) $\rho(z) = z^2 - 2z + 1 = (z - 1)^2$. Here $z = 1$ is not a simple root, it is repeated and, since it has magnitude equal to 1, the method is not zero stable.

Example 02

Find the roots of the first characteristic polynomial for the linear multistep scheme

$$y_{n+2} - 2y_{n+1} + y_n = h\left(f_{n+2} + 2f_{n+1} + f_n\right)$$

and hence determine whether or not the scheme is zero stable.

Consistency

- A scheme that is zero stable will produce approximations that do not grow in size in a way that is not present in the exact, analytic solution.
- Zero stability is a required property, but it is not enough on its own.
- There remains the issue of whether the approximations are close to the exact values.
- The truncation error of the general linear multistep method is a measure of how well the differential equation and the numerical method agree With each other.

It is defined by

$$\tau_j = \frac{1}{\beta}\left(\frac{c_0}{h}y(jh) + c_1 y'(jh) + c_2 h y''(jh) + c_3 h^2 y'''(jh) + \dots\right) = \frac{1}{\beta h}\sum_{p=0}^{\infty} c_p h^p y^{(p)}(jh)$$

where $\beta = \sum \beta_j$ is a normalising factor.

It is the first few terms in this expression that will matter most in what follows, and it helps us that there are formulae for the coefficients which appear

$$c_0 = \sum \alpha_j, \quad c_1 = \sum (j\alpha_j - \beta_j), \quad c_2 = \sum \left(\frac{j^2}{2}\alpha_j - j\beta_j \right), \quad c_3 = \sum \left(\frac{j^3}{3!}\alpha_j - \frac{j^2}{2}\beta_j \right)$$

and so on, the general formula for $p \geq 2$ is $c_p = \sum \left(\frac{j^p}{(p)!}\alpha_j - \frac{j^{p-1}}{(p-1)!}\beta_j \right)$.

Recall that the truncation error is intended to be a measure of how well the differential equation and its approximation agree with each other We say that the numerical method is consistent with the differential equation if τ_i tends to zero as h $\longrightarrow$ 0.

The linear multistep scheme is said to be consistent if $c_0 = 0$ and $c_1 = 0$

Example 01

Show that Euler's method $(y_{n+1} = y_n + hf_n)$ is consistent.

Solution

In this case $\alpha_1 = 1$, $\alpha_0 = -1$, $\beta_1 = 0$ and $\beta_0 = 1$. It follows that

$$c_0 = \sum \alpha_j = 1 - 1 = 0 \qquad \text{and} \qquad c_1 = \sum j\alpha_j - \beta_j = 1\alpha_1 - (\beta_0 + \beta_1) = 1 - (1 + 0) = 0$$

and therefore Euler's method is consistent.

Example 02

Show that the trapezium method $(y_{n+1} = y_n + \frac{h}{2}(f_{n+1} + f_n))$ is consistent.

Order

A linear multistep method is said to be of **order** p if

$$c_0 = c_1 = c_2 = \cdots = c_p = 0 \qquad \text{and} \qquad c_{p+1} \neq 0$$

Example 01:

Find the order of

(a) Euler's method

(b) The trapezium method.

(a) We have already found that $c_0 = c_1 = 0$ so the first quantity to calculate is

$$c_2 = \sum \left(\frac{j^2}{2} \alpha_j - j\beta_j \right) = \tfrac{1}{2}\alpha_1 - \beta_1 = \tfrac{1}{2}$$

which is not zero and therefore Euler's method is of order 1. (Or, in other words, Euler's method is first order.)

(b) We have already found that $c_0 = c_1 = 0$ so the first quantity to calculate is

$$c_2 = \sum \left(\frac{j^2}{2} \alpha_j - j\beta_j \right) = \tfrac{1}{2}\alpha_1 - \beta_1 = \tfrac{1}{2} - \tfrac{1}{2} = 0$$

this is equal to zero, so we must calculate the next coefficient

$$c_3 = \sum \left(\frac{j^3}{3!} \alpha_j - \frac{j^2}{2} \beta_j \right) = \tfrac{1}{6}\alpha_1 - \tfrac{1}{2}\beta_1 = \tfrac{1}{6} - \tfrac{1}{4} = -\tfrac{1}{12}$$

which is not zero. Hence the trapezium method is of order 2 (that is, it is second order).

Example 02:

Find the order of the 4-step, explicit linear multistep scheme

$$y_{n+4} - y_{n+3} = \frac{h}{24}\left(55f_{n+3} - 59f_{n+2} + 37f_{n+1} - 9f_n\right)$$

In the established notation we have $\alpha_4 = 1$, $\alpha_3 = -1$, $\alpha_2 = 0$, $\alpha_1 = 0$ and $\alpha_0 = 0$. The β terms similarly come from the coefficients on the right hand side (remembering the denominator of 24).
Now

$$c_0 = \sum \alpha_j = 0 \qquad \text{and} \qquad c_1 = \sum j\alpha_j - \beta_j = 0$$

from which we conclude that the method is consistent.
We also find that

$$c_2 = \sum \tfrac{1}{2}j^2\alpha_j - j\beta_j = 0, \qquad c_3 = \sum \tfrac{1}{6}j^3\alpha_j - \tfrac{1}{2}j^2\beta_j = 0,$$

$$c_4 = \sum \tfrac{1}{24}j^4\alpha_j - \tfrac{1}{6}j^3\beta_j = 0 \qquad c_5 = \sum \tfrac{1}{120}j^5\alpha_j - \tfrac{1}{24}j^4\beta_j = 0.348611 \quad \text{to 6 d.p.}$$

(The exact value of c_5 is $\frac{251}{720}$.)

Because c_5 is the first non-zero coefficient we conclude that the method is of order 4.

Key Point

The numerical approximation to the initial value problem converges to the actual solution as $h \to 0$ if

1. the scheme is zero stable
2. the scheme is consistent